A

PROJECT REPORT

ON

A Digital Walkie Talkie Architecture Utilizing GFSK

Submitted For

EC256:
Electronics and Communication Laboratory

Submitted to

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Chapter 1

Abstract

This project details the design and implementation of a digital half duplex communication system functioning as a wireless walkie talkie within the 2.4 GHz Industrial, Scientific, and Medical (ISM) band. The architectural workflow initiates with the acquisition of an analog acoustic signal via an electret microphone, which undergoes active pre-amplification before being digitized by a microcontroller based Analog to Digital Converter (ADC). The discrete data stream is subsequently transmitted via an NRF24L01 radio frequency module employing Gaussian Frequency Shift Keying (GFSK) modulation. At the receiving node, the incident payload is demodulated and synthesized back into an analog waveform utilizing a high frequency Pulse Width Modulated (PWM) Digital to Analog Converter (DAC). A passive low pass filter attenuates high frequency quantization noise before final power amplification drives an electrodynamic speaker. To validate the hardware implementation, the underlying mathematical principles of the GFSK modulation and demodulation processes, alongside sampling and quantization theories, are rigorously modeled and verified using MATLAB. The system provides a robust framework for understanding discrete signal processing and digital radio frequency transmission tailored for localized environments.

Chapter 2

Introduction

The project details the design and implementation of a digital half-duplex communication system, transitioning from an initial analog FM concept to a discrete digital architecture operating in the 2.4 GHz ISM band. Functioning as a wireless walkie-talkie prototype, it captures, conditions, and digitally samples acoustic energy, then broadcasts the data using Gaussian Frequency Shift Keying (GFSK). This end-to-end communication system practically demonstrates theoretical baseband electronics, including sampling, quantization, digital modulation, and signal reconstruction, up to practical radio frequency (RF) transmission.

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2.1 Objectives

1. To design and construct a functional digital half duplex voice communication link capable of operating independently of commercial cellular or Wi-Fi networks.
2. To implement active analog signal conditioning techniques to prepare microvolt level acoustic signals for microcontroller based Analog to Digital Conversion (ADC).
3. To utilize GFSK modulation for robust wireless data transport within the 2.4 GHz ISM band.
4. To reconstruct continuous analog waveforms from discrete digital payloads using high frequency Pulse Width Modulation (PWM) and passive low pass filtering.
5. To mathematically model and verify the discrete sampling, quantization, and GFSK modulation processes using MATLAB, comparing theoretical expectations with physical hardware outputs.

2.2 Components Used

- **Microcontroller (Arduino UNO):** Serves as the central processing unit, executing the ADC algorithms at the transmitter and the PWM based DAC synthesis at the receiver.

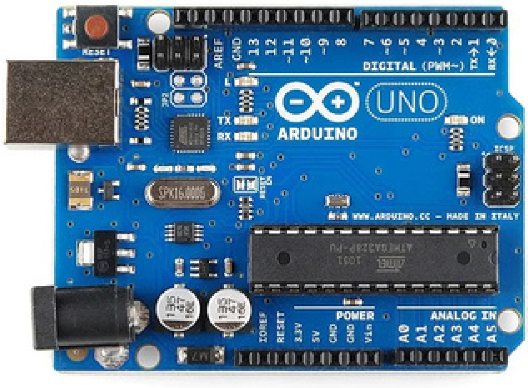

Figure 2.1: Arduino UNO

RF Transceiver (NRF24L01): The primary wireless module responsible for managing the physical layer, executing GFSK modulation and demodulation at 2.4 GHz.

Figure 2.2: RF Transceiver

Electret Microphone: The primary acoustic transducer used to capture human voice inputs.

Figure 2.3: Electret Microphone

Operational Amplifier (LM358): Deployed as an active pre amplifier to boost the weak microphone signals to an optimal voltage range for digital sampling.

Figure 2.4: Operational Amplifier

Audio Power Amplifier (LM386): An integrated circuit utilized to provide the necessary current and voltage gain to drive the output transducer.

Figure 2.5: Audio Power Amplifier

Electrodynamic Speaker (8 Ohm, 0.5 W): The output transducer that converts the final reconstructed electrical waveform back into audible acoustic waves.

Figure 2.4: Electrodynamic Speaker

Passive Components: Various precision resistors and capacitors utilized to bias the active components and construct the essential RC low pass filter for signal reconstruction.

Chapter 3

Design and Implementation

3.1 System Overview:

The proposed half-duplex communication prototype constitutes an integrated architecture engineered for the wireless transmission and reception of human voice signals. The operational pipeline captures analog acoustic waveforms, digitizes them for radio frequency transmission over the 2.4 GHz Industrial, Scientific, and Medical (ISM) band, and subsequently reconstructs the audio baseband signal at the receiver terminal.

The signal acquisition phase initiates with an electret microphone, whose output is amplified by an LM358 operational amplifier. The integrated Analog-to-Digital Converter (ADC) of the ATmega328P microcontroller subsequently digitizes this amplified baseband signal. This discrete-time data is transferred to the NRF24L01 transceiver module, which employs Gaussian Frequency Shift Keying (GFSK) modulation to broadcast the payload via radio frequency packets.

At the receiving node, a symmetrical hardware configuration captures the incident electromagnetic waves. The recipient NRF24L01 module demodulates the GFSK carrier, extracting the digital payload and forwarding it to the microcontroller. The receiving Arduino functions as a discrete Digital-to-Analog Converter (DAC) by generating a high-frequency Pulse Width Modulated (PWM) signal to reconstruct the analog waveform. This synthesized signal traverses a passive low-pass filter to attenuate high-frequency quantization and switching artifacts, undergoes power amplification via an LM386 module, and is finally transduced into acoustic waves by an 8-ohm electrodynamic speaker.

3.2 Key Functional Block:

3.2.1 Audio Input and Pre-Amplification Stage

- **Components:** Electret Microphone, LM358 Operational Amplifier.
- **Functionality:** Unamplified audio signals generated by the electret microphone possess low amplitude, typically in the millivolt range, rendering them susceptible to quantization error if sampled directly. The LM358 operational amplifier functions as an active pre-amplifier stage, providing the necessary voltage gain to elevate the weak baseband signal to the dynamic range required by the microcontroller's ADC.

3.2.2 Central Processing Unit (Arduino UNO)

- **Components:** ATmega328P Microcontroller.
- **Transmitter Role:** The microcontroller samples the amplified analog baseband signal via its internal ADC, converting the continuous-time voltage into discrete digital representations. This digitized payload is sequentially clocked out to the RF transceiver via the Serial Peripheral Interface (SPI) protocol.
- **Receiver Role:** Upon receiving the digital payload from the demodulating RF module, the microcontroller synthesizes an analog output utilizing a high-frequency PWM peripheral, thereby functioning as an effective DAC to restore the continuous waveform.

3.2.3 RF Transceiver Module (NRF24L01)

- **Operating Principle:** Operates within the 2.4 GHz ISM spectrum utilizing GFSK modulation.
- **Functionality:** This module governs the wireless physical layer. It executes the modulation of the digital baseband data stream onto a high-frequency carrier wave for transmission and performs the inverse demodulation process on incident RF signals to recover the original bitstream at the receiving terminal.

3.2.4 Audio Output and Power Amplification Stage

- **Components:** Passive RC Low-Pass Filter, LM386 Audio Amplifier IC, 8-Ohm 0.5-Watt Speaker.
- **Functionality:** The unconditioned PWM output generated by the microcontroller constitutes a discrete square wave rich in high-frequency harmonics. A passive resistor-capacitor (RC) low-pass filter is implemented to attenuate the carrier frequency, thereby extracting the fundamental audio envelope. Because this filtered voltage signal lacks the sufficient current-driving capability required for an inductive load, the LM386 audio amplifier provides the requisite power gain to drive the 8-ohm speaker transducer effectively.

Chapter 4

Operations

The operational pipeline of the digital transceiver represents a complete cycle of signal acquisition, digital conversion, RF transmission, and acoustic reconstruction. The system executes the following discrete sequential phases:

Phase 1: Signal Acquisition and Pre Amplification The operation commences when the user speaks into the electret microphone. The microphone acts as a transducer, converting varying acoustic pressure into corresponding electrical voltage fluctuations. Because these baseband signals exist in the low millivolt range, they possess insufficient amplitude to be accurately sampled. The signal is routed through an LM358 operational amplifier configured in a non inverting amplifier topology. This stage provides the necessary voltage gain, mapping the acoustic waveform to a 0 to 5V dynamic range suitable for the microcontroller.

Phase 2: Analog to Digital Conversion (ADC) The amplified continuous time signal enters an analog input pin on the transmitting Arduino UNO. The internal ADC samples the waveform at a predetermined frequency governed by the Nyquist Shannon sampling theorem to prevent aliasing. The continuous voltage levels are quantized into discrete digital values (typically utilizing an 8 bit resolution to optimize processing speed and transmission bandwidth). This digital payload effectively represents the acoustic envelope in a binary format.

Phase 3: Digital Modulation and RF Transmission The microcontroller sequentially clocks the digitized audio payload to the NRF24L01 module via the Serial Peripheral Interface (SPI) bus. Upon receiving the data packet, the NRF24L01 module executes Gaussian Frequency Shift Keying (GFSK) modulation. GFSK smooths the digital pulses using a Gaussian filter before frequency modulating the 2.4 GHz carrier wave. This Gaussian smoothing reduces the spectral bandwidth of the transmitted signal, minimizing adjacent channel interference. The modulated RF packets are then broadcast through the module's integrated antenna.

Phase 4: RF Reception and Demodulation Operating in a synchronized listening state, the receiving NRF24L01 module captures the incoming 2.4 GHz electromagnetic waves. The module demodulates the GFSK carrier, extracting the original discrete binary sequence. It performs a cyclic redundancy check (CRC) to ensure data integrity and then forwards the verified digital payload to the receiving Arduino UNO via its local SPI bus.

Phase 5: Digital to Analog Conversion (DAC) Synthesis Because the Arduino UNO lacks a dedicated hardware DAC, it synthesizes the analog signal using a high frequency Pulse Width Modulation (PWM) technique. The microcontroller reads the incoming digital values and dynamically adjusts the duty cycle of a square wave output on a dedicated digital pin. A high data value generates a wider pulse (higher average voltage), while a low data value generates a narrower pulse (lower average voltage), directly mirroring the amplitude of the original sampled waveform.

Phase 6: Low Pass Filtering and Power Amplification The raw PWM signal is a digital square wave containing severe high frequency switching harmonics that are unsuitable for audio playback. The signal is routed through a passive resistor capacitor (RC) low pass filter. The cutoff frequency of this filter is mathematically calculated to attenuate the PWM carrier frequency while permitting the baseband audio frequencies to pass, effectively integrating the square wave back into a smooth continuous waveform. Finally, because the filtered signal lacks the current capacity to move a speaker cone, it is fed into an LM386 audio power amplifier. The LM386 boosts the overall power of the signal, which subsequently drives the 8 Ohm speaker, successfully reproducing the original spoken message.

Chapter 6

Code Implementation

6.1 Mathematical Verification of the Process using MATLAB

- Code file

6.2 Receiver End Code

- Code file

6.3 Transmitter End Code

- Code file

Chapter 5

Results and Conclusion

5.1 Results

1. The prototype successfully facilitated the acquisition, wireless transmission, and acoustic reconstruction of human voice signals between two transceiver nodes operating on the 2.4 GHz ISM band.
2. Theoretical mathematical models simulating the GFSK modulation, discrete sampling, and quantization processes were successfully executed in MATLAB, exhibiting a high degree of correlation with the physical hardware behavior.
3. The integration of an active LM358 pre amplification stage, coupled with an RC low pass filter at the DAC output, effectively mitigated quantization noise and high frequency switching artifacts to yield intelligible audio fidelity.
4. The system maintained a stable radio frequency link within the specific operational range of the NRF24L01 transceiver operating in its low power state, ensuring synchronous real time communication.

5.2 Applications

1. Subterranean and Industrial Communication: Instantaneous tactical communication deployments in mines or dense industrial factories characterized by severe radio frequency attenuation and a complete lack of cellular network infrastructure.
2. Structural Intercom Systems: Secure, localized half duplex intercom architectures for institutional buildings or residential facilities requiring instant connectivity.
3. Off Grid Acoustic Monitoring: Short range acoustic monitoring systems operating entirely independently of commercial Wi-Fi or mobile networks, constrained efficiently to the low power operational range of the incorporated RF transceiver.

5.3 Conclusion

This project successfully validates fundamental digital communication theorems through both physical hardware implementation and MATLAB based mathematical modeling. By synthesizing analog signal conditioning, microcontroller driven discrete data processing, and high frequency GFSK modulation via the NRF24L01, the architecture effectively bridges theoretical baseband electronics with practical radio frequency transmission. The finalized prototype satisfies the primary objectives of establishing a reliable, low latency audio link specifically optimized for instantaneous communication in isolated environments devoid of standard cellular coverage, thereby demonstrating the practical viability of embedded digital communication systems.

Chapter 6

Cost Structure

Component	Quantity	Estimated Unit Price (INR)	Estimated Total Price (INR)
Arduino UNO (Compatible Board)	2	₹200	₹400
NRF24L01 Transceiver Module	2	₹120	₹240
Electret Microphone	1	₹25	₹25
LM358 Operational Amplifier IC	1	₹15	₹15
LM386 Audio Amplifier Module	1	₹80	₹80
8 Ohm 0.5 W Speaker	1	₹40	₹40
Resistors, Capacitors, and Diodes (Assorted)	1 Set	₹50	₹50
General Purpose Transistors (e.g., BC547/2N2222)	1 Set	₹30	₹30
Power Supply (9V Battery with snap)	1	₹25	₹25
Total Estimated Cost			₹905